

Harsh Jha

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EDUCATION

VIT Vellore — B.Tech (Bachelor of Technology) in Computer Science and Engineering *Expected Graduation: July 2028*

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Operating Systems

PROFILE SUMMARY

Computer Science and Engineering undergraduate seeking an Engineering & Technology internship, with hands-on experience completing technology projects that deliver measurable outcomes and business improvements: a ~3x runtime speedup, ~25% lower RMSE, and ~40% fewer data errors. GSoC 2026 Contributor coached by research mentors through code reviews and design discussions, and Full Stack Developer Intern who learned an enterprise platform and its internal tooling quickly. Collaborates in cross-functional agile teams, follows engineering standards and procedures, and presents technical results clearly to stakeholders.

EXPERIENCE

GSoC Contributor — QC-Devs (Google Summer of Code 2026)

May 2026 – Aug 2026

- Engineered core modules of a Python-based electrostatics engine for production use, researching unfamiliar numerical systems independently while receiving coaching and mentoring through design discussions and code reviews with research mentors in an agile workflow.
- Debugged and optimized production numerical algorithms, reducing RMSE by ~25% and delivering a ~3x runtime speedup through object-oriented refactoring and performance profiling.
- Authored 30+ automated unit tests through high-quality pull requests, raising test coverage from ~60% to ~90% and applying coding standards and review procedures to keep code reproducible and production-ready with minimal defects.
- Delivered 4+ merged contributions to a multi-person open-source project with minimal guidance, learning new systems quickly and reporting progress to mentors on a regular cadence.

Full Stack Developer Intern — CCIEExpert

July 2026

- Built a Python CLI policy-analysis tool and a Manifest V3 browser extension (OAuth2 authentication, HTTP request handling, rate-limited API ingestion) integrating with Cisco Secure Access, taking the project from independent research and systems design through production deployment.
- Learned a new enterprise security platform and its APIs quickly while collaborating with a small cross-functional agile team on requirements gathering, code reviews, and testing; gave and received technical feedback and presented results to stakeholders.
- Debugged and resolved defects across the extension's HTTP/API integration layer prior to production release, ensuring minimal defects in shipped code.

PROJECTS

MoneyMetrics | [Link](#) — Python, React, JavaScript, HTML/CSS, Firebase

- Architected and shipped a full-stack finance dashboard end-to-end, designing the system independently and integrating the Gemini API for automated Z-score anomaly detection across user transaction data to deliver data-driven insights.
- Built reconciliation logic and test coverage that cut data-inconsistency errors by ~40%; deployed and actively maintained in production on Netlify, handling HTTP-based API requests with real user-facing traffic.

Ffprime | [Link](#) — Python, NumPy, SciPy

- Contributed object-oriented refactors via reviewed pull requests to an established multi-person open-source scientific computing project (theochem/ffprime), restructuring core electrostatics modules for maintainability.
- Vectorized numerical routines with NumPy for a ~3x performance speedup and authored 32 unit tests, ensuring strong test coverage and minimal defects in production code.

ACHIEVEMENTS & CERTIFICATIONS

- Google Big Code 2026 — Qualifier cleared, ranked top 15,000 nationally.
- Selected as GSoC 2026 Contributor (QC-Devs) — competitive, merit-based selection process.
- Certifications: Infosys Springboard (Machine Learning, Deep Learning, AI, GenAI); Python for Data Science, AI & Development — IBM (Coursera).

SKILLS

Languages: Java, Python, C++, JavaScript, SQL

Web / Front-End: HTML, CSS, JavaScript, React

Practices: Systems Design, Cross-Functional Collaboration, Object-Oriented Design, Agile/Scrum, Project Delivery, Debugging, Unit Testing, Pull Requests, Code Review, Technical Communication, RESTful APIs, HTTP, OAuth2

Data & Scientific Computing: NumPy, SciPy, Numerical Methods, Data Analysis, Performance Profiling

Databases & Tools: SQL, Oracle SQL, Firebase, Git